

B03 Gate B Protected Review - Education Enrollment

Protected diagnostic review - preserve fresh A/B/C evidence

Protected review - saved genuine Gate attempt required

Use this only after the matching genuine Gate attempt is saved. Diagnose, repair, close the review, then produce changed evidence. Later Gates remain state-machine protected.

Objective source controls

students 30; courses 8; enrollments 60; attendance 180; assessments 90; student-program rows 35

Metric	Protected control
Completed enrollments	45
Assessment score total	6770.00
Naive attendance x assessment joined rows	270
Naive repeated score total	20300.00
Score inflation	13530.00

Why a plausible model can still be wrong

- enrollments, attendance events and assessments are separate processes/grains. Joining the two child facts through enrollment repeats assessment scores.
- student_programs is a legitimate membership bridge; five students have multiple program memberships.
- Course and Student dimensions can be conformed context, but fact-to-fact flattening is not required to compare them.

Reasoning and evidence rubric

- State each business process and exact one-row grain before assigning fact/dimension/bridge roles.
- Prove dimension/business-key uniqueness and state foreign-key/cardinality expectations.
- Separate facts at different grains. Predict and test fan-out before any cross-fact aggregation.
- Use bridges only for legitimate membership; never as a broom for duplicate dimension defects.
- Keep role-playing dates explicit and name the question each date role answers.
- Reconcile raw counts/totals to modeled outputs. Unexplained differences remain visible blockers.
- Provide an implementation-ready handoff plus a rejected alternative and why it fails.

Critical failures

- Mixed grain or hidden fan-out.
- Arbitrary many-to-many/DISTINCT repair.
- Target or transaction measures repeated at a lower/higher incompatible grain.
- Failed reconciliation hidden or explained only by “the visual looks right.”
- Cannot reproduce/defend the model contract.

Repair and changed retry route

- Save the original failed/developing Gate attempt.
- Open only this matching protected review after the saved attempt exists.
- Repair the named weak competency using the targeted remediation cards.
- Close this review and produce a changed micro-proof in a different small scenario.
- Save the changed evidence. Gate B/C still require the existing explicit fresh full-retest assignment when applicable.

Next route

PASS: B03 ends. FAIL/developing: second repair; Gate C stays sealed.